

Experiment	Sieve Analysis Test
Objective	To determine the particle size distribution of a given soil sample.
Apparatus	Set of sieves including the pan and cover, Sieve shaker, Balance (with an accuracy to 0.1g)
Theory	The particle size distribution of a given soil sample is indicated by the grading curve. The graph of cumulative % finer (passing) vs log particle size is known as the grading curve. The particle sizes corresponding to 10%, 30% and 60% finer (D_{10}, D_{30} and D_{60}) obtained from the grading curve are used to calculate the Coefficient of Curvature (C_C) and Uniformity Coefficient (C_U) values. These values indicate the size distribution or the grading of the soil particles in the sample. $C_U = \frac{D_{60}}{D_{10}} \qquad C_C = \frac{(D_{30})^2}{D_{10} \times D_{60}}$ Where D_{10} is called the Effective Size for a well graded soil: $C_U > 5$ for a well graded soil: $1 < C_C < 3$

Procedure

1. Record the mass of the soil sample.
2. Clean the set of sieves.
3. Record the mass of each sieve.
4. Arrange the set of sieves in the descending order of sieve sizes.
5. Place the soil sample on the top sieve and close the lid.
6. Conduct the sieving operation by using the sieve shaker.
7. Weigh and record the mass of soil retained on each sieve.

Calculations

1. calculate the % of soil retained on each sieve
2. calculate the cumulative % of soil retained
3. calculate the cumulative % of soil passing (% finer)
4. plot the grading curve (% finer vs particle size) on the semi log graph sheet
5. Calculate the C_c , C_u and D_{10} values.

Results

C_c , C_u and D_{10} values

Conclusion

Identify the soil type using the grading curve.

Observations

Sieve analysis test

Name -----

Reg Number -----

Course -----

Date -----

Sieve Size (mm)	Mass of sieve (g)	Mass of sieve + soil retained (g)	Mass of soil retained (g)	% retained	Cumulative % retained	Cumulative % passing